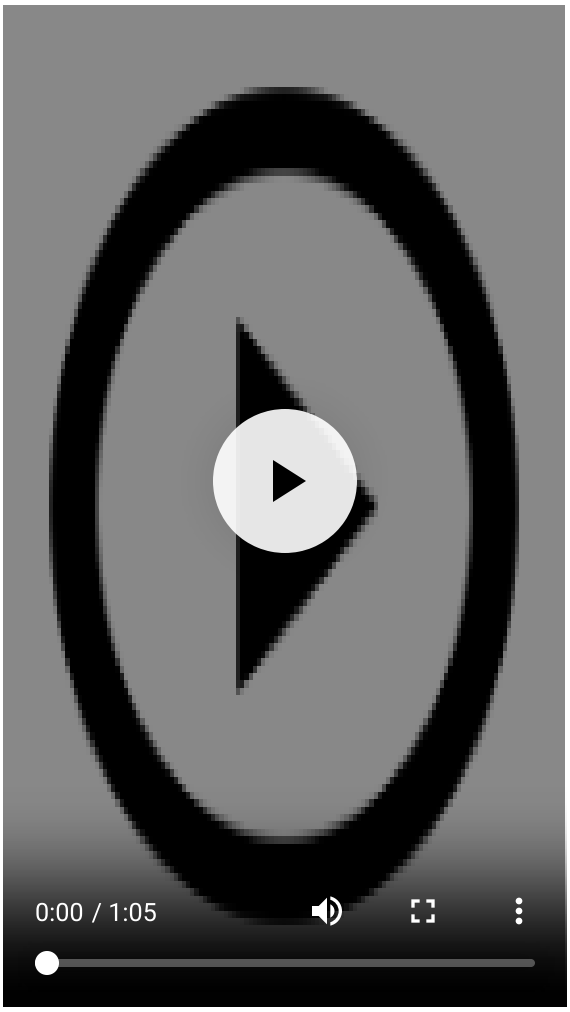
Activity 4.2: Develop the Flutter Mobile Application



By the end of this activity, your group should have a working mobile application that is ready to be integrated with the ASP.NET web application.

Supporting Video: Building a Flutter Application

Watch the relevant sections of the official Flutter video:

Building your first Flutter App - with a Codelab! (https://www.youtube.com/watch?v=8sAyPDLorek)

Video section	What to focus on
14:08	Adding interactive elements

54:42	Implementing application functionality
1:02:44	Navigation
1:23:03	Adding another page

The example in the video is different from your stakeholder project. Use it to understand the Flutter development process, then apply what you learn to your own approved interfaces.

Additional support: Use Supabase with Flutter | Supabase Docs (https://supabase.com/docs/guides/getting-started/quickstarts/flutter)explains how to connect a Flutter application to Supabase.

Activity at a Glance

You will develop	**Main outcome**
Flutter project structure	Organised mobile source code
Approved mobile screens	Working user interfaces
Forms and navigation	Users can move through the application
Required mobile functions	Features based on the SRS
Supabase connection	Mobile application uses the shared backend
Initial function checks	Evidence of working mobile features

What You Must Do

Step 1: Review the Approved Mobile Application Requirements

Open your updated SRS and review the mobile interface designs, use case diagram and sequence diagrams.

Identify the functions that must be implemented in Flutter.

Complete the following table for your project.

Mobile function	**Related requirement or use case**	**Required screen**

Short example: Salon appointment system

Mobile function	**Related use case**	**Required screen**
Sign in	Login	Login screen
Request appointment	Request Appointment	Appointment form
View appointment status	View Appointment Status	My Appointments screen

Your group's implementation must follow your own approved requirements, not the salon example.

Step 2: Prepare the Flutter Project

Open the group's Flutter project or create it if it has not yet been prepared.

Keep the mobile application in the mobile/ folder of the group's GitHub repository. Use meaningful names for screens, models and services.

Check that the application runs in your development environment before implementing the remaining functions.

Step 3: Develop the Approved Screens

Use the wireframes from Unit 3 to build the main mobile screens.

For each screen, implement the required interface elements.

Interface element	**Example**
Text field	Customer name
Date or time input	Preferred appointment date
Button	Submit Request
List	My Appointments
Status display	Pending, Approved or Rejected

Your interfaces must be clear, consistent and appropriate for the intended users.

Step 4: Implement Navigation and Forms

Connect the screens so users can move through the mobile application.

Add appropriate form validation. For example, a user should receive a helpful message when a required field is empty.

Short example: A customer opens the appointment form, enters the required details and selects Submit Request. The application checks the information before attempting to save it.

The exact navigation and validation must follow your project's requirements.

Step 5: Connect Flutter to the Shared Supabase Backend

Use the same Supabase project configured in Activity 4.1.

Install and configure the supabase_flutter package, initialise the client and connect the required application functions to the approved database tables. Follow the official quickstart for the current setup steps.

Use Supabase with Flutter (https://supabase.com/docs/guides/getting-started/quickstarts/flutter)

Flutter Client Library (https://supabase.com/docs/reference/dart/introduction)

Implement the database operations required by your mobile application.

Operation	**Salon example**
Create	Save a new appointment request
Read	Retrieve the customer's appointments
Update	Change information where the user is authorised to do so
Delete	Remove a record only where the requirement and permissions allow it

Important: Use the appropriate client publishable key and correctly configured database permissions. Do not include Supabase secret or service-role keys in the mobile application or GitHub repository.

Step 6: Implement the Main Mobile Functions

Implement the mobile functions identified in Step 1.

Where authentication is required, connect the sign-in process to Supabase Authentication and make sure users can access only the information and functions permitted by their role.

For each function, check that the user can complete the intended process from beginning to end.

Step 7: Perform Initial Function Checks

Use suitable dummy data to check that the mobile application works.

Function or check	Result	Action required
Application launches successfully		
Main screens display correctly		
Navigation works		
Forms validate input		
Mobile application connects to Supabase		
Required records can be saved		
Required records can be retrieved		
Authentication works, where required		
Access restrictions work as intended		

Fix problems identified during development. The complete prototype will undergo formal manual testing in Unit 5.

Step 8: Commit Your Work and Update Project Records

Commit the developed mobile application to the group's GitHub repository.

Update the existing Microsoft Project file and Project Progress Tracker to show completed tasks, responsible group members, progress and any challenges.

If an approved requirement or design needed to change during implementation, record the change in the SRS.

What to Keep as Evidence

Evidence	What it should show
Mobile application screenshots	Implemented screens and navigation
Function screenshots or short recordings	Main mobile functions working
Supabase evidence	Required information saved or retrieved through the mobile application
GitHub commits	Development progress and contributions

Updated Microsoft Project	Current development progress
Updated Project Progress Tracker	Completed tasks, contributions and challenges

Do not include passwords, secret keys or confidential personal information in your evidence.

Activity Outcome

A working Flutter mobile application with the approved screens, navigation and main functions implemented.

The mobile application must connect to the shared Supabase backend and be ready for integration with the ASP.NET web application.

Before Moving On

Confirm that the group has:

- implemented the approved mobile screens and navigation;
- developed the required forms and mobile functions;
- connected Flutter to the shared Supabase backend;
- checked the main functions using suitable test data;
- retained evidence of the working mobile application;
- committed relevant code to GitHub; and
- updated Microsoft Project and the Project Progress Tracker.

Next: Activity 4.3: Develop the ASP.NET Web Application.